

CHARTS AS EPISTEMIC ACTIONS: EXPERIENCE-DRIVEN MULTIMODAL ANALYSIS IN DATA AGENTS

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Paper under double-blind review

ABSTRACT

Charts are important cognitive tools that help human analysts identify trends, patterns, and anomalies. Although multimodal large language models (MLLMs) can interpret charts, the analytical utility of charts for data-analysis agents remains underexplored. In this work, we first conduct a probe study based on annotated questions to examine models’ ability to interpret evidence from charts and their chart-use behavior in open-ended autonomous analysis. We find that the evaluated models effectively extract useful evidence from charts and tables, but differ substantially in whether they actively use charts during autonomous analysis. Moreover, charts and text exhibit task-dependent complementarity. These findings suggest that charts should not serve merely as post-hoc presentation tools, but should form an intrinsic part of data-analysis agents’ reasoning. Building on this insight, we propose **Co-evolving Analysis and Representation Experiences**. CARE contrasts analysis trajectories with different evidential representations to jointly evolve analytical and representational experience. This approach helps agents learn how to construct, select, and use chart-based evidence while mitigating modality-use biases. Experiments on three held-out benchmarks show that CARE encourages models to actively use charts for analysis and significantly outperforms baseline methods.

1 INTRODUCTION

Data agents are intelligent systems centered on large language models (LLMs) that interpret analytical goals, access data environments, invoke analytical tools, and produce data-grounded insights. Recent progress in ReAct-style reasoning–acting loops and tool use (Yao et al., 2023) has strengthened autonomous planning, code execution, and iterative analysis, moving data agents from proof-of-concept demonstrations toward practical assistants capable of complex tasks—a promising path for business decision-making, scientific inquiry, and data-science automation.

Nevertheless, most data agents still conduct analysis primarily in text. Models generate Text-to-SQL queries or Python scripts from a schema and advance reasoning from textualized execution results. Human analysts rely on charts to recognize trends, distributions, anomalies, and relationships among variables; they externalize candidate patterns visually and then select further computation and verification accordingly. Why has this step not yet been fully integrated into the data-agent analytical loop? More generally, *in what form should intermediate analytical evidence and insights be presented so that information is preserved and subsequent reasoning is supported?* The question concerns not only whether to use charts, but also how analytical results should be organized, represented, and consumed.

Prior work on charts follows three broad lines: generation, understanding, and representation selection. Generation methods produce and refine visualizations from data and user intent, improving data fidelity and chart quality (Dibia, 2023; Yang et al., 2024; 2025; Namgoong et al., 2025; Lu et al., 2026; Wang et al., 2025). Understanding methods strengthen interpretation through structure recovery, numerical reasoning, and visual tools (Masry et al., 2023; Han et al., 2023; Xu et al., 2023; Liu et al., 2023a;b; Zhang et al., 2024; Kaur et al., 2026; Das et al., 2025). Multimodal representation research compares textual and visual encodings of the same information and shows that representation choice affects reasoning outcomes (Zhou et al., 2025; Ho et al., 2026; Xing et al., 2026; Lin et al., 2026). These strands typically target specified chart generation, interpretation of externally supplied charts,

or selection among preset representations. They have not systematically measured the benefit of charts in *open-ended* data analysis, nor answered whether data agents can productively exploit open plotting capabilities or which kinds of analytical evidence charts versus text should carry.

To assess the practical utility of charts for data agents, we construct PROBETASK from synthetic analysis trajectories and examine chart-reading competence, behavior under open plotting tools, and the performance of charts and text across analytical tasks and cognitive demands. Two main findings emerge. First, evaluated models can extract useful evidence from charts, yet they do not reliably convert access to plotting tools into consistent analytical gains—a pronounced *capability–utility* gap. Second, charts and text serve as task-dependent, complementary evidence representations: charts are better suited to trends, patterns, and anomalies, whereas text excels at precise reading, conditional verification, and integrating multiple pieces of evidence. These results indicate that how intermediate evidence is represented directly shapes analytical reasoning; because utility varies with task and analytical stage, representation strategy should be learned from trajectories and outcome feedback rather than reduced to fixed task-to-modality rules.

Motivated by this view, we propose CARE (Co-evolving Analysis and Representation Experiences), which jointly evolves analytical and representational experience. For each task, CARE samples multiple analysis trajectories and compares their final outcomes to distill two kinds of reusable knowledge: what analytical evidence to construct, and how to present and use that evidence. The two experience types are stored in separate repositories but updated and version-selected jointly according to task performance under their combined effect. At inference time, the data agent retrieves relevant experiences given the current task, analytical operation, and intermediate results, and dynamically decides subsequent analysis steps and whether to use text, charts, or a hybrid—without hand-crafted modality-routing rules. We evaluate CARE on several external data-analysis benchmarks held out from experience evolution, and use ablations with single experience types, chart disabling, and non-evolved experience to test the incremental value of dual-experience co-evolution and visual representation.

In summary, our main contributions are as follows:

- We systematically probe the analytical utility of charts for data agents on synthetic trajectories (PROBETASK), revealing a gap between chart understanding and productive use of open plotting, and summarizing task-dependent complementarity between charts and text as evidence representations.
- We propose CARE (Co-evolving Analysis and Representation Experiences), which jointly evolves analytical and representational experience from outcome-scored trajectories so that agents learn both what evidence to construct and how to represent and use it, without hand-crafted modality-routing rules.
- Experiments on three external data-analysis benchmarks (DDR-Bench, DAComp-DA, and DS Bench) with six LLM backbones demonstrate that CARE consistently outperforms matched baseline agents; ablations on single experience types, chart disabling, and non-evolving experience quantify the contribution of dual-experience co-evolution and visual representation.

2 CHARTS IN THE ANALYTICAL LOOP

In this section, we construct PROBETASK from synthetic analysis trajectories and evaluate data agents in a realistic tool-use loop with open plotting. We first describe the task suite, metrics, and agent settings (§2.1), then analyze chart-reading competence, autonomous modality use, and the complementary roles of text and charts (§2.2).

2.1 EXPLORATORY PROBE DESIGN

Probe construction. Our goal is to isolate how representations of intermediate results affect analytical reasoning, rather than to introduce another end-to-end data-analysis benchmark. This requires tasks that pair raw tables and open-ended questions with executable analyses and reference insights. InsightBench (Sahu et al., 2025) provides these elements over enterprise tables and therefore supports controlled interventions on the agent’s analytical interface. We take its first 50 source-data

Table 1: Interface ablation on PROBETASK with gpt-5.6-sol: scores for each of the four agent settings.

Agent configuration	Score
No-tool	29.4
Text-only	77.2
Chart-only	78.4
Free-style	77.6

Table 2: Free-style on PROBETASK across six LLM backbones: score; modality use (# tasks); mean ReAct rounds and tokens.

LLM backbone	Score	Used modality (# tasks)			ReAct cost	
		Text	Chart	Both	Rounds	Tokens
Claude Opus 5	83.1	43	0	157	3.9	17,401
Claude Sonnet 5	79.2	102	0	98	4.0	18,724
GPT-5.6 Luna	76.2	198	0	2	3.1	6,835
GPT-5.6 Sol	77.6	198	0	2	2.9	6,392
DeepSeek V4.1 Flash	80.2	143	0	57	6.2	30,051
HY3	73.1	198	0	2	3.8	7,725

flags and split each associated insight unit into one ProbTask. This process yields 219 candidates. After excluding 17 tasks with empty questions, we retain 202 evaluable tasks from 49 flags.

Each ProbTask contains a raw CSV table, a dataset description, an analytical question, a reference analysis, and a target insight. The agent receives the table and question, together with the description, schema, and first three rows as initial context. The target insight remains hidden and is used only for evaluation. The agent must select and execute the required operations, such as filtering, feature derivation, aggregation, sorting, temporal transformation, or statistical testing, and return a direct natural-language answer.

Task annotation. We use two complementary taxonomies for post-hoc analysis. *Question type* specifies the answer sought by a task: numeric lookup, composition, rank or extreme, group comparison, temporal change, relationship, anomaly or risk, diagnosis, or prediction. *Cognitive load* specifies the primary operation required to derive the answer from an intermediate analytical view: locate, scan and select, compare, recognize a pattern, interpret a relation, integrate dimensions, or perform conditional reasoning. An LLM first assigns each label from the task evidence, after which a human annotator reviews and confirms it. The resulting labels support only post-hoc slice analysis; they never define hand-crafted modality-routing rules.

Metrics. A common GPT-5.6 Sol judge scores each answer against the question and target insight on a 0–4 scale. We map normalized judge scores to $[0, 100]$ and report their mean. Slice analyses use task-paired differences with flag-clustered bootstrap intervals (Efron & Tibshirani, 1993) and randomization tests.

Agent settings. All settings share the same vanilla ReAct agent with one general-purpose Bash tool. Each call starts a fresh process, while files persist in a shared working directory and the CSV is exposed read-only. We hold the task, model, initial context, interaction limit, and tool budget fixed across groups; only the analytical interface differs:

- **No-tool:** no execution tool; the agent answers from the description, schema, and three example rows only.
- **Text-only:** tool output exposes stdout and stderr; plotting is prohibited.
- **Chart-only:** stdout is hidden; the agent receives generated pixel images and stderr, with analytical results explicitly prohibited in stderr.
- **Free-style:** stdout, stderr, and pixel images are all returned; the agent may use text, charts, or both.

Experiments. On the full PROBETASK suite ($n=202$), we run an *interface ablation* with GPT-5.6 Sol under all four settings above, and a separate *model comparison* with six backbones—Claude Opus 5, Claude Sonnet 5, GPT-5.6 Luna, GPT-5.6 Sol, DeepSeek V4.1 Flash, and HY3—under the Free-style interface. The same GPT-5.6 Sol judge scores every run. In §2.2, we analyze these results in depth.

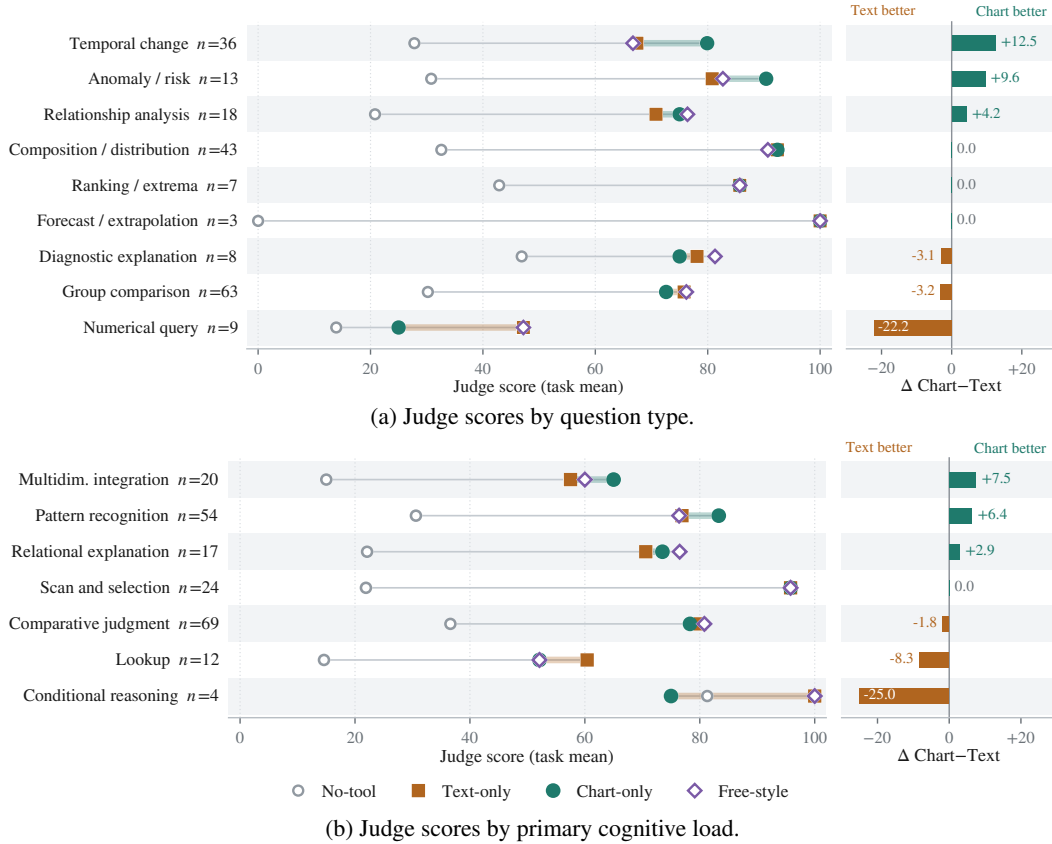


Figure 1: Slice analysis of the four agent settings on PROBETASK ($n=202$). Markers show slice means; the right column gives Chart-Text Δ (faded when $n < 10$). The bottom row of a aggregates all tasks.

2.2 ANALYSIS AND FINDINGS

We now analyze the probe runs and summarize three sets of results: aggregate interface effects, model-specific modality use under Free-style, and slice-level complementarity between text and charts.

LLMs can conduct analysis through charts. Table 1 reports mean scores for the four agent settings. GPT-5.6 Sol scores only 29.4 without tools, whereas its scores increase to 77.2 with text and 78.4 with charts. The chart condition therefore achieves performance comparable to the text condition and improves over the no-tool condition by 49.0 points. These results show that multimodal LLMs can extract sufficient analytical evidence from charts to answer data-analysis questions.

Open-ended modality choice exhibits strong model-specific bias. Table 2 compares Free-style trajectories across six LLMs, including modality-use patterns and interaction cost. Marked variation appears under the same interface. Claude Opus 5 combines text and charts on 157 tasks, while Claude Sonnet 5 and DeepSeek V4.1 Flash do so on 98 and 57 tasks, respectively. In contrast, GPT-5.6 Luna, GPT-5.6 Sol, and HY3 use both modalities on only 2 tasks each and rely on text alone on the remaining 198. Chart-use frequency also has no monotonic relationship with performance: DeepSeek V4.1 Flash uses charts less often than Claude Sonnet 5 but attains a higher score, while the three low-frequency models obtain different scores despite identical use counts. Thus, access to the same multimodal interface does not produce a common modality-selection strategy; open-ended modality choice remains strongly shaped by model-specific priors.

The advantage of charts is concentrated in change and pattern recognition. Figure 1 breaks down the four settings by question type and cognitive load. Charts score 0.799 on temporal-change questions, compared with 0.674 for text, yielding a gain of 0.125. Charts also lead by 0.096 on anomaly/risk questions and by 0.042 on relationship-analysis questions. The cognitive-load results exhibit the same pattern: charts score 0.833 on pattern recognition, exceeding text by 0.064, and lead by 0.029 on relational explanation and by 0.075 on multidimensional integration. In contrast, the two modalities perform similarly on composition/distribution, ranking/extrema, and scan-and-selection tasks. Free-style performs below the chart condition on both temporal change and pattern recognition, indicating that access to both modalities does not automatically preserve the advantages of visual representation.

Text is better suited to exact reading and conditional verification. Figure 1 shows that text scores 0.472 on numerical queries, compared with 0.250 for charts, and scores 0.604 on lookup tasks, compared with 0.521 for charts. Text also has smaller advantages of 0.032 on group comparison and 0.018 on comparative judgment. On conditional reasoning, text scores 1.000 while charts score 0.750, although this slice contains only four tasks and therefore provides only exploratory evidence. Overall, the relative advantage of text is concentrated in tasks that require exact reading, information location, and conditional verification rather than in data-analysis tasks generally.

Taken together, these results reveal a gap between multimodal capability and effective multimodal analysis. Although current LLMs can extract useful evidence from self-generated charts, merely providing plotting tools does not ensure that agents will use them when beneficial or coordinate them with textual evidence. Their behavior remains strongly shaped by model-specific modality preferences, while the relative utility of text and charts varies across analytical demands. The central limitation is therefore not chart understanding alone, but the absence of a mechanism that jointly determines what analytical view to construct and how that view should be represented and used. Because these decisions depend on the task and the evolving analysis state, they cannot be reduced to a fixed task-to-modality routing rule. This motivates learning coupled analysis and representation strategies from completed trajectories and outcome feedback, as developed in the next section.

3 LEARNING TO USE CHARTS IN AGENTIC DATA ANALYSIS

We present an experience-driven method that teaches data agents to use charts within the process of agentic data analysis. The method distills analysis experience and representation experience from trajectories generated under different representation conditions. The validated experiences then guide the agent in selecting analytical operations and evidence representations, allowing charts to support intermediate reasoning rather than serve only as final outputs.

We first describe the structure and contents of the experience package. We then present conditioned trajectory rollouts, trajectory scoring and attribution, and the iterative procedure for experience distillation and validation. Finally, we explain how the resulting experience package guides downstream analysis. Figure 2 provides an overview.

3.1 EXPERIENCE PACKAGE

3.1.1 STORAGE AND TOOL LAYERS.

The experience package consists of a storage layer and a tool layer. The storage layer uses a file-system structure in which an entry document organizes independent experience documents. Each document contains metadata fields such as `name`, `description`, and `category`, together with its experience content, which supports direct reading, editing, and versioning.

The tool layer provides `Read`, `Create`, `Edit`, and `Bash` for reading, creating, modifying, and managing experience documents. The builder agent uses all four tools to maintain the package, whereas the data agent accesses experience only through `Read`. Experience is loaded progressively. The data agent first receives document metadata and then reads relevant content according to the current task and analysis state.

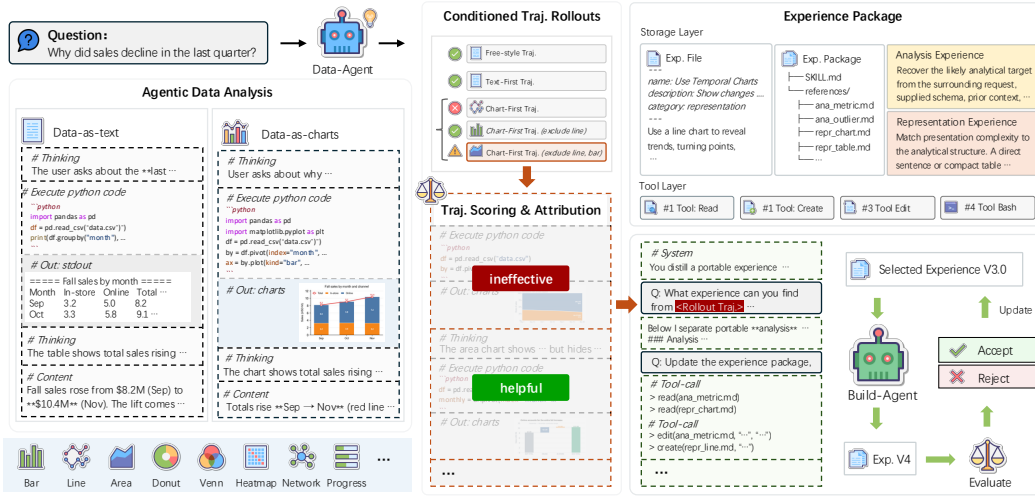


Figure 2: Overview of chart use in agentic data analysis and CARE. **Left:** a data agent answers analytical questions from tables using text or chart evidence under open plotting. **Center:** conditioned trajectory rollouts are scored and attributed to separate helpful from ineffective analysis and representation choices. **Right:** analysis and representation experiences are stored in an experience package, iteratively updated by a build agent, and used to guide downstream analysis.

3.1.2 ANALYSIS AND REPRESENTATION EXPERIENCES.

We extract two types of experience. Analysis experience $\mathcal{E}_{\text{ana}}$ summarizes how to select, organize, and verify analytical operations. Representation experience $\mathcal{E}_{\text{repr}}$ summarizes when to use text, charts, or hybrid representations and how to use them to advance subsequent analysis. The two types are stored separately and updated and validated as one experience package. We denote its v -th version by $\mathcal{E}^{(v)} = (\mathcal{E}_{\text{ana}}^{(v)}, \mathcal{E}_{\text{repr}}^{(v)})$.

3.2 CONDITIONED TRAJECTORY ROLLOUTS

We denote the i -th PROBE TASK by $p_i = (D_i, q_i, z_i, y_i^*)$, where the four elements represent the data, question, task context, and reference answer. For each task, we generate multiple analysis trajectories under different representation conditions while holding the base agent, tools, and interaction budget fixed. The trajectory generated under condition ρ is denoted by τ_i^ρ . Each task yields a bundle of five trajectories,

$$\mathcal{B}_i = \{\tau_i^U, \tau_i^T, \tau_i^{C_1}, \tau_i^{C_2}, \tau_i^{C_3}\}. \quad (1)$$

Here, τ_i^U is an unconstrained rollout, τ_i^T is a text-preferred rollout, and $\tau_i^{C_k}$ for $k \in \{1, 2, 3\}$ are three chart-preferred rollouts. These conditions change only the representation preference and do not restrict the analytical operations available to the agent.

To increase diversity among chart trajectories, each later chart-preferred rollout excludes chart types used in preceding rollouts. Let $\mathcal{K}_i^{(k)}$ denote the set of chart types used by the k -th chart-preferred rollout. We impose

$$\mathcal{K}_i^{(k)} \cap \left(\bigcup_{j < k} \mathcal{K}_i^{(j)} \right) = \emptyset, \quad k \in \{2, 3\}. \quad (2)$$

This constraint increases trajectory diversity without assuming that charts outperform text or that any chart type is intrinsically superior. The five trajectories from the same PROBE TASK form the basic unit for subsequent scoring and experience distillation.

3.3 TRAJECTORY SCORING AND ATTRIBUTION

For each trajectory, a trajectory judge produces an outcome score and step-level attribution using the task question and reference answer y_i^* . The outcome score evaluates the final answer. The attribution labels key steps as `Helpful`, `Ineffective`, or `Harmful`, and diagnoses the contributions of both analytical operations and representation choices. Together, these signals indicate which behaviors the builder agent should retain or avoid.

3.4 EXPERIENCE DISTILLATION AND VALIDATION

We split the experience-construction and validation sets by `PROBETASK`, ensuring that all trajectories from the same task remain in the same split. At each iteration, we sample b tasks from the experience-construction set and provide their complete trajectory bundles, outcome scores, and step-level attributions to the builder agent.

Each update has two steps. During insight induction, the builder agent compares trajectories and their attributions to identify transferable analytical and representational patterns. During experience consolidation, it reads the current package and merges these patterns into existing documents or creates new analysis and representation experiences.

The builder agent edits the previously accepted package $\mathcal{E}^{(v-1)}$ and produces a candidate $\tilde{\mathcal{E}}^{(v)}$. Let $Q(\mathcal{E})$ denote validation performance with experience package $\mathcal{E}$. The update rule is

$$\mathcal{E}^{(v)} = \begin{cases} \tilde{\mathcal{E}}^{(v)}, & Q(\tilde{\mathcal{E}}^{(v)}) > Q(\mathcal{E}^{(v-1)}), \\ \mathcal{E}^{(v-1)}, & \text{otherwise.} \end{cases} \quad (3)$$

The candidate is accepted only when validation performance strictly improves. After multiple iterations, the final package $\mathcal{E}^*$ is frozen for downstream tasks.

3.5 EXPERIENCE-GUIDED ANALYSIS

For downstream tasks, the data agent uses the frozen package $\mathcal{E}^*$. The system first provides document metadata, after which the agent reads relevant experiences according to the question and current analysis state. Analysis experience guides the selection and verification of analytical operations. Representation experience guides whether intermediate results use text, charts, or both. The base model parameters remain unchanged. The external experience package guides chart use according to the task and analysis state, making charts part of the analytical process rather than a fixed preference or final presentation artifact.

4 EXPERIMENTS

The planned main evaluation spans three benchmarks with different task forms and official protocols: long-horizon autonomous exploration, full analytical report generation, and exact-answer data-analysis question answering. At the time of writing, CARE results are not complete; accordingly, we specify the design and leave unfilled cells in Section 4.3 explicitly marked as TBD.

4.1 BENCHMARKS

DDR-Bench. Long-horizon exploratory analysis: it evaluates sustained exploration without predefined queries (Liu et al., 2026). We use the public, reproducible 10-K setting over a SQLite database of SEC financial disclosures for 100 companies. The official LLM-as-checker protocol reports Message-Wise and Trajectory-Wise accuracy and their mean.

DAComp-DA. Analysis and report generation: it contains 100 open-ended tasks over potentially multi-table, large SQLite databases and requires a complete report with analysis, conclusions, and visualizations (Lei et al., 2025). We report Completeness, Accuracy, Conclusiveness, Readability, Professionalism, and Visualization. Official Overall is 60% rubric correctness, 10% readability, 10% professionalism, and 20% visualization, rather than an unweighted six-metric average.

Table 3: Results on *DDR-Bench* benchmark.

Method	LLM	Msg. (%)	Traj. (%)	Overall (%)
<i>Publicly reported benchmark scores</i>				
ReAct	Claude Sonnet 4.5	77.61	60.58	69.10
ReAct	GPT-5.1	37.12	44.25	40.69
ReAct	GPT-5.2	44.89	41.09	42.99
ReAct	DeepSeek V3.2	60.08	38.15	49.12
ReAct	GLM-4.6	60.31	36.02	48.17
ReAct	Kimi K2	51.06	30.84	40.95
ReAct	MiniMax M2	44.17	26.88	35.53
Baseline	Claude Sonnet 5	61.2	52.2	56.7
	GPT-5.6 Luna	44.5	45.1	44.8
	GPT-5.6 Sol	60.2	58.5	59.4
	DeepSeek V4.1 Flash	64.2	59.0	61.6
	HY4 Preview	TBD	TBD	TBD
CARE	Claude Sonnet 5	TBD	TBD	TBD
	GPT-5.6 Luna	TBD	TBD	TBD
	GPT-5.6 Sol	TBD	TBD	TBD
	DeepSeek V4.1 Flash	TBD	TBD	TBD
	HY4 Preview	TBD	TBD	TBD

Table 4: Results on *DSBench* benchmark.

Method	LLM	Task (%)	Comp. (%)	Overall (%)
<i>Publicly reported benchmark scores</i>				
Human	—	64.06	67.33	65.70
Model-only	GPT-3.5	20.39	11.85	16.12
Model-only	GPT-4o	28.11	19.26	23.69
Code Inter.	GPT-3.5	11.16	8.23	9.70
Code Inter.	GPT-4o	23.82	22.65	23.24
AutoGen	GPT-3.5	20.82	13.80	17.31
AutoGen	GPT-4o	34.12	26.72	30.42
Baseline	Claude Sonnet 5	88.47	88.42	88.45
	GPT-5.6 Luna	75.11	73.84	74.47
	GPT-5.6 Sol	89.9	89.4	89.7
	DeepSeek V4.1 Flash	TBD	TBD	TBD
	HY4 Preview	92.70	91.19	91.95
CARE	Claude Sonnet 5	TBD	TBD	TBD
	GPT-5.6 Luna	TBD	TBD	TBD
	GPT-5.6 Sol	TBD	TBD	TBD
	DeepSeek V4.1 Flash	TBD	TBD	TBD
	HY4 Preview	TBD	TBD	TBD

DSBench. Exact-answer data analysis: it derives end-to-end tasks from real ModelOff competitions (Jing et al., 2024). Its Data Analysis subset has 40 challenges and approximately 466 questions, with inputs that may include spreadsheets, text, and images. We report Task-level Accuracy, competition-macro-averaged Competition-level Accuracy, and their mean.

Isolation. PROBETASK is used for probing, experience evolution, and development gating only. DDR-Bench, DAComp-DA, and DSBench are independent external tests held out from experience evolution. Splits use the natural source unit—company, task, or challenge—to prevent data-source overlap.

4.2 EXPERIMENTAL SETUP

Baseline. The evaluation contains reported benchmark agents, a Baseline Agent reproduced in our common harness, and CARE. Baseline and CARE use the same ReAct loop, Bash/Python/SQL tools, stdout, stderr, and chart feedback; only CARE retrieves the frozen DEL. Thus improvement cannot be attributed merely to access to plotting.

LLMs. The planned backbones are Claude Opus 5, Claude Sonnet 5, GPT-5.6 Luna, GPT-5.6 Sol, DeepSeek V4.1 Flash, and HY3. Within each backbone, Baseline and CARE share model version, decoding, system instructions, data access, and interaction budget.

Experience evolution. All DEL learning is performed on PROBETASK only, using the rollout, outcome-only judge, and development-set gating procedure in Section 3.4. Each backbone evolves its experience libraries independently from the same initial structure; the accepted version is frozen before evaluation on DDR-Bench, DAComp-DA, and DSBench.

Each benchmark uses its official evaluator: DDR-Bench’s GPT-5-mini checker, DAComp-DA’s official rubric/GSB suite, and DSBench’s answer-correctness procedure. Methods share samples, judge prompts, and parameters within each benchmark. Reported prior results are directly compared only when splits and protocols match. All core comparisons are paired, with clustered bootstrap confidence intervals at the company, task, and challenge levels, respectively.

4.3 MAIN RESULTS

Cells marked “TBD” denote runs or official evaluations that are not yet complete; they are placeholders, not estimates. We report each benchmark separately and do not average scores across benchmarks with incompatible scales.

DDR-Bench. Table 3 reports long-horizon exploratory analysis on the reproducible 10-K setting.

DSBench. Table 4 reports exact-answer data analysis on the Data Analysis subset.

Table 5: Results on *DAComp-DA* benchmark.

Method	LLM Backbone	Completeness	Accuracy	Insightfulness	Readability	Analytical Depth	Visualization	Overall
<i>Publicly reported benchmark scores</i>								
OpenHands	GPT-5	60.98	40.3	49.39	35.51	69.80	21.40	46.99
OpenHands	DeepSeek-V3.1	49.88	33.25	41.66	36.00	33.20	11.00	33.87
OpenHands	Qwen3-Coder	33.42	21.21	25.06	20.00	13.73	4.80	24.28
DA-Agent	GPT-5	64.23	43.81	56.89	43.59	76.80	27.44	50.84
DA-Agent	DeepSeek-V3.1	48.74	32.97	42.43	37.25	35.00	11.45	34.33
DA-Agent	Qwen3-Coder	35.12	20.05	25.53	19.37	13.42	5.15	25.13
Baseline	Claude Sonnet 5	70.27	52.55	68.85	90.80	89.00	62.80	69.84
	GPT-5.6 Luna	66.21	46.61	65.45	83.83	80.74	7.66	54.32
	GPT-5.6 Sol	71.74	58.04	72.42	94.60	93.00	61.80	73.24
	DeepSeek V4.1 Flash	83.91	67.80	81.77	98.0	94.44	75.72	82.23
	HY4 Preview	78.55	60.38	74.39	96.74	94.78	66.52	79.40
CARE	Claude Sonnet 5	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	GPT-5.6 Luna	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	GPT-5.6 Sol	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	DeepSeek V4.1 Flash	TBD	TBD	TBD	TBD	TBD	TBD	TBD

We will report Message-Wise accuracy of $[X.X]$ and Trajectory-Wise accuracy of $[Y.Y]$ for CARE, corresponding to $[\Delta X]$ and $[\Delta Y]$ point improvements over Baseline. If Trajectory-Wise gains dominate, experience primarily benefits long-horizon exploration and final evidence synthesis; if both metrics rise together, CARE improves both local insight discovery and end-to-end trajectory coverage.

DAComp-DA. Table 5 reports analysis and report generation with full analytical reports.

We will track official Overall (from $[A]$ to $[B]$) together with Completeness and Accuracy (changes $[\Delta C]$ and $[\Delta A]$) to assess whether gains reflect substantive evidence and task coverage rather than surface formatting, and Visualization ($[\Delta V]$) to assess whether visual presentation improves. We interpret visual representation as serving analysis only when content-correctness metrics and Visualization improve jointly.

We will report Task-level Accuracy $[X.X]$ and competition-macro Competition-level Accuracy $[Y.Y]$, with Baseline-relative gains of $[\Delta X]$ and $[\Delta Y]$ points. Concurrent improvements on both metrics would indicate that gains are not driven solely by competitions with many tasks; divergent changes would motivate inspection of large-competition dominance.

Cross-benchmark synthesis. Across dataset-backbone pairs, we expect CARE to outperform the matched Baseline on $[X/Y]$ combinations, suggesting that jointly learning analytical operations and evidence representations transfers beyond task forms closely aligned with PROBETASK—to long-horizon exploration, full report generation, and exact-answer analysis. If effect sizes differ markedly across benchmarks, we will relate them to task openness, required analytical depth, and output modality rather than pooling raw scores. Finer task slices, ablations, and mechanism analyses will be reported once the corresponding runs are complete; the main text will report aggregate effects with the clustered bootstrap intervals defined above.

4.4 PLANNED ABLATIONS AND MECHANISM ANALYSIS.

The ablations compare analysis experience only, representation experience only, charts disabled, and non-evolving experiences against full CARE. Trajectory analysis records whether a chart triggers a new analytical operation, whether visual candidates receive numerical verification, whether charts appear only as final output, and whether unnecessary or repeated plotting occurs. A representative case will contrast a long-text failure, correct aggregation without useful representation, and a CARE trajectory that discovers a turning point visually and verifies its magnitude textually. These traces explain behavior but do not substitute for benchmark outcomes.

5 RELATED WORK

5.1 DATA-ANALYSIS AGENTS.

LLM-driven data-analysis agents increasingly follow a plan–execute–observe–revise paradigm. DS-Agent couples language-model reasoning with an executable analysis environment and retrieves historical cases to support data-science problem solving; Data Interpreter organizes complex tasks as dynamic directed acyclic graphs, advancing analysis through hierarchical decomposition and code execution; AIDE models machine-learning solution development as tree search in code space; DS-STAR handles heterogeneous data-analysis tasks via iterative planning and plan-sufficiency verification; and AutoKaggle and related systems adopt multi-agent division of labor for data understanding, solution planning, program development, and result review (Guo et al., 2024; Hong et al., 2025; Jiang et al., 2025; Nam et al., 2025; Li et al., 2025). Meanwhile, DeepAnalyze targets end-to-end coverage of data preparation, analysis, modeling, visualization, and report generation through specialized training, and InsightBench extends evaluation to open-ended, multi-step business insight generation (Zhang et al., 2025; Sahu et al., 2025). Although these lines of work continue to advance planning, search, collaboration, and training mechanisms, their analysis loops still rely predominantly on textual representations of structured data and on code-execution outputs as intermediate feedback. Some systems can generate charts, but few isolate whether agents inspect self-generated charts and leverage the visual patterns they reveal to form hypotheses and steer subsequent analysis. We focus on how visual evidence enters the intermediate reasoning loop of data-analysis agents.

5.2 VISUAL MODALITIES IN DATA SCIENCE.

In parallel with data-analysis agents, research on visual modalities has made substantial progress in chart generation, understanding, and representational choice. For chart generation, LIDA established a pipeline from data summaries and visualization goals to plotting code; MatPlotAgent uses rendered visual feedback to iteratively repair visualization programs; ChartMimic, AMACE, and MultiVis-Agent further improve reliability through cross-modal reconstruction, multi-agent collaboration, and iterative optimization; and ChartOptimiser shows that chart design should adapt to the analytical task at hand (Dibia, 2023; Yang et al., 2024; 2025; Namgoong et al., 2025; Lu et al., 2026; Wang et al., 2025). For chart understanding, UniChart, ChartLlama, and ChartBench advance chart reading and reasoning via specialized training and large-scale evaluation; DePlot, MatCha, and TinyChart improve numerical analysis through derendering, structural recovery, and programmatic reasoning; and ChartAgent and Charts-of-Thought demonstrate that visual tools combined with structured reasoning can strengthen complex chart analysis (Masry et al., 2023; Han et al., 2023; Xu et al., 2023; Liu et al., 2023a;b; Zhang et al., 2024; Kaur et al., 2026; Das et al., 2025). Building on this foundation, FRES, Format Matters, and TableDART study effectiveness differences and dynamic selection among textual, image, and chart representations, while systems such as EvidFuse construct visual evidence on demand during analysis or reporting (Zhou et al., 2025; Ho et al., 2026; Xing et al., 2026; Lin et al., 2026). However, existing work typically generates user-specified charts, interprets externally provided charts, or selects among pre-existing representations; it has not systematically studied data-analysis agents that start from raw structured data, jointly decide analytical operations and evidential representations, and use self-generated charts as intermediate cognitive tools for discovering patterns and advancing subsequent analysis.

6 CONCLUSION

We study how charts can serve as internal epistemic actions rather than end-of-analysis illustrations. PROBETASK shows that current LLMs can complete end-to-end analysis through self-generated charts, yet often fail to turn plotting-tool access into consistent gains, and exhibit strong model-specific biases when text and charts are both available. Task slices indicate complementary roles: charts support trends and pattern discovery, whereas text better preserves precise, verifiable evidence. CARE jointly evolves analysis and representation experiences from outcome-scored trajectories, learning both what evidence to construct and how to represent and use it without hand-crafted modality rules. Experiments on held-out benchmarks indicate that this co-evolution improves over matched baselines, supporting the view that representation choice is part of the analytical policy rather than a cosmetic output.

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A COMPLETE PROBETASK SLICE RESULTS

Tasks may have multiple analytical-operation labels, while each task has one primary cognitive-operation label. Low- n rows are included for completeness and should not be interpreted as stable modality effects.

Table 6: Scores by analytical operation.

Slice	n	No-tool	Text	Chart	Free-style
Filtering	109	0.257	0.752	0.752	0.755
Feature derivation	116	0.257	0.739	0.752	0.744
Aggregation	163	0.305	0.773	0.791	0.781
Normalization	21	0.143	0.571	0.512	0.583
Sorting	22	0.284	0.818	0.830	0.784
Alignment/joining	8	0.188	0.719	0.563	0.719
Temporal change	37	0.237	0.757	0.838	0.757
Distributional stats.	1	1.000	1.000	1.000	1.000
Statistical inference	10	0.200	0.925	0.900	0.900
Forecasting	3	0.000	1.000	1.000	1.000

B ITEMS REQUIRED BEFORE SUBMISSION

The final manuscript still requires: confirmation that CARE and PROBETASK do not conflict with prior names; final probe and framework artwork; a qualitative trajectory figure; pseudocode for training and online inference; completed external-benchmark and ablation results with statistical tests; judge robustness and human-agreement checks; representative experience entries and full prompts; and compute, cost, and implementation details.

Table 7: Scores by primary cognitive operation.

Slice	n	No-tool	Text	Chart	Free-style
Lookup	12	0.146	0.604	0.521	0.521
Scan and selection	24	0.219	0.958	0.958	0.958
Comparative judgment	69	0.366	0.801	0.783	0.808
Pattern recognition	54	0.306	0.769	0.833	0.764
Relational explanation	17	0.221	0.706	0.735	0.765
Multidimensional integration	20	0.150	0.575	0.650	0.600
Conditional reasoning	4	0.813	1.000	0.750	1.000